

Math100C - WK4

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2026.04.24

1.

Identify (up to isomorphism) all finite groups G which have exactly three irreducible characters of dimensions 1, 2, and 3, respectively. (You may use the *Sylow theorems*, which you can look up if they weren't covered in your 100A.)

2.

Let G be a finite cyclic group. Let S be the set of generators of G . Let χ be any character of G (not necessarily irreducible).

1. Show that

$$\sum_{g \in S} \chi(g)$$

is an integer. (*Hint*: reduce to the irreducible case.)

2. Show that

$$\prod_{g \in S} |\chi(g)|^2$$

is a *nonnegative* integer.

3. Show that if $\chi(g) \neq 0$ for all $g \in S$,

$$\sum_{g \in S} |\chi(g)|^2 \geq |S|.$$

(*Hint*: combine the previous part with the arithmetic-geometric mean inequality.)

3.

Let G be a finite group. Let χ be an irreducible character such that $\chi(g) \neq 0$ for all $g \in S$. Use the previous exercise to deduce that

$$\sum_{g \in G \setminus \{1\}} |\chi(g)|^2 \geq |G| - 1$$

and deduce that $\chi(1) = 1$. In other words, an irreducible character χ of dimension greater than 1 must take the value 0 somewhere; this theorem is originally due to Burnside.

4.

Let G be a finite group. Prove that the following two conditions are equivalent.

1. The entries of the character table of G are all integers.
2. Any two elements of G which generate the same cyclic subgroup of G belong to the same conjugacy class.

(*Hint:* use orthogonality to show that any two elements of G which have the same value on every character are conjugate.) Then deduce as a corollary that for every positive integer n , the character table of S_n contains only integers.